

Dr. Awadhesh Sharma

Department of Civil Engineering, IIT Roorkee, India

+91-8005702360 | asharma@ce.iitr.ac.in | ce.0905400028@gmail.com

Google Scholar (h-index: 4, Citations: 47+) | ORCID: 0000-0002-0375-0443

EXECUTIVE SUMMARY

Ph.D. scholar and Post-Doctoral Researcher at IIT Roorkee with specialized expertise in **Fracture Mechanics**, **Fatigue Analysis**, and **Ultra-High Performance Concrete (UHPC)**. My research integrates advanced experimental techniques—including **Digital Image Correlation (DIC)** and **Acoustic Emission (AE)**—with computational modeling (**Abaqus**, **COMSOL**) to solve complex structural challenges.

EDUCATION

Indian Institute of Technology (IIT), Roorkee

Ph.D. in Civil Engineering

Roorkee, India

Oct 2024

Thesis: Mechanical and Fracture Studies on Ultra-High Performance Fibre Reinforced Concrete.

CGPA: 7.857/10.0

Motilal Nehru National Institute of Technology (MNNIT)

M.Tech. in Civil Engineering

Allahabad, India

2013 – 2015

Thesis: Acid effect on Fibre Reinforced GGBS Concrete.

CGPA: 8.050/10.0

Uttar Pradesh Technical University (UPTU)

B.Tech. in Civil Engineering

Lucknow, India

2009 – 2013

Percentage: 72.22% (First Division)

RESEARCH & PROFESSIONAL EXPERIENCE

Indian Institute of Technology (IIT), Roorkee

Post-Doctoral Fellow (MoRTH Sponsored Project)

Roorkee, India

Oct 18, 2025 – Present

Project: Design parameters for Ultra-High-Performance Fibre Reinforced Concrete (UHPFRC) in bridge infrastructure.

Key Contribution: Developing new techniques for designing UHPFRC and establishing code-level design parameters.

Mentoring Ph.D. scholars in AE sensor deployment and fatigue testing protocols.

Indian Institute of Technology (IIT), Roorkee

Post-Doctoral Fellow

Roorkee, India

July 21, 2025 – Oct 17, 2025

Conducted numerical simulation and damage analysis of concrete composites.

Facilitated laboratory operations and contributed to ongoing departmental consultancy.

Indian Institute of Technology (IIT), Roorkee

Research Associate (NBCC Sponsored Project)

Roorkee, India

Dec 2024 – July 20, 2025

Project: Ultra-High-Performance Concrete for Building Applications: Development and Design Parameters.

Responsible for mix design optimization and mechanical characterization for high-rise structures.

JECRC University

Assistant Professor

Jaipur, India

Sept 2015 – July 2017

Taught Structural Analysis and Concrete Technology; mentored 10+ UG thesis projects.

PUBLICATIONS

Refereed Journals (SCI/Scopus)

- B. Kumar, A. Sharma, S. Ray, "Characterization of crack-bridging and size effect on ultra-high performance fibre reinforced concrete under fatigue loading," *International Journal of Fatigue*, vol. 182, 108158, Elsevier, 2024.
- B. Kumar, A. Sharma, S. Ray, "An analytical approach for characterizing the fracture behaviour of ultra-high-performance fibre reinforced concrete," *Composite Structures*, vol. 332, 117922, Elsevier, 2024.
- A. Sharma, S. Ray, M.A. Iqbal, "Physico-chemical, Mechanical and Fracture Characterization of Sustainable Ultra-high Performance Composite," *Magazine of Concrete Research*, ICE Publishing, 2024.

4. B. Kumar, **A. Sharma**, S. Ray, "Characterization of the fracture behavior of UHPFRC by considering the influence of specimen size and fiber content," *Procedia Structural Integrity*, **Elsevier**, 2024.
5. **A. Sharma**, S. Ray, M.A. Iqbal, "Characterization of fracture development in geometrical similar UHPC beams using acoustic emission (AE) technique," *Materials Today: Proceedings*, **Elsevier**, 2023.

Papers Under Review

1. V.W. Masih, G. Ruiz, S.T. Maganty, **A. Sharma**, S. Ray, V. Vishwakarma, A. De La Rosa, R.C. Yu, "Modified Brazilian Test for Studying Crack Initiation and Fracture Behavior in SFRC Using DIC & AE," *Construction and Building Materials*, **Elsevier** (Under Review).

Conference Proceedings (International)

1. **A. Sharma**, S. Ray, M.A. Iqbal, "Damage analysis of ultra-high performance fibre reinforced concrete through acoustic emission under fatigue loading," *3rd International Conference on Construction Materials and Structures (ICCMS-2025)*, IIT Tirupati, India, July 3-5, 2025.
2. **A. Sharma**, S. Ray, M.A. Iqbal, "Fracture Behaviour of Ultra-High Performance Concrete," *FraMCoS-X*, Bayonne, **France**, 2019.
3. **A. Sharma**, et al., "Compression Dispersion Efficiency of Concrete Struts using Stone Dust," *RTCCE-2014*, MNNIT Allahabad, Dec 2014.
4. **A. Sharma**, et al., "Correlation between Non-Destructive Strength and Destructive Strength of Recycled Aggregate Concrete Struts," *RTCCE-2014*, MNNIT Allahabad, Dec 2014.

Conference Proceedings (National)

1. **A. Sharma**, "Influence of Constituent Materials on the Compressive Strength of Ultra-High-Performance Concrete," *National Conference on Innovative Construction Machineries, Materials and Methods (ICMMMID-2026)*, Dehradun, India, Feb 9-10, 2026.

TECHNICAL SKILLS ACADEMIC SERVICE

Technical Skills: Abaqus, COMSOL Multiphysics, STAAD.Pro, MATLAB, AutoCAD, Origin Pro, L^AT_EX.

Experimental Skills: DIC (Digital Image Correlation), Acoustic Emission (AE), Fatigue Testing, Servo-Hydraulic Machines (MTS).

Reviewer: *Sādhanā* (Springer), *Construction and Building Materials* (Elsevier - Assistance).

Awards: **Best Paper Award** (ICCMS-2022) | **Int. Travel Grant** (France, 2019) | **MHRD Fellowship** (2017-2022).

Sports: Winner in Kabaddi, 19th Josh J. Lal Sports Festival (MNNIT Allahabad, 2014).

WORKSHOPS, TRAINING VOLUNTEERING

- **Volunteer Coordinator:** Int. Conf. on Advances in Construction Materials (ACMS-2018), IIT Roorkee.
- **Short Term Course:** Advances in Structural Engineering (ASE-2021), NIT Hamirpur.
- **FDP (2 Weeks):** Soil-Structure Interactions and Its Applications (SSIA-2020), GB Pant Institute.
- **FDP (TEQIP-III):** Recent Advances and Innovations in Design of Structures (RAIDS-2020).
- **Course (TEQIP-II):** Earthquake Resistant Design & Detailing (ERDSS-2014), MNNIT Allahabad.
- **Workshop:** Extreme Loading on Structures (2017), IIT Roorkee & CSIR-CBRI.
- **Internship:** UP Rajkiya Nirman Nigam Ltd. Lucknow (July 2012) & UP PWD Gorakhpur (June 2012).

REFERENCES

Dr. Sonalisa Ray
Professor, Civil Engineering, IIT Roorkee
✉ sonalisaray@gmail.com

Dr. Mohd. Ashraf Iqbal
Professor, Civil Engineering, IIT Roorkee
✉ ashraf.iqbal@ce.iitr.ac.in